

The Kauffman bracket of a link diagram D has axioms

$$\langle \emptyset \rangle = 1; \quad \langle \bigcirc \sqcup D \rangle = (q + q^{-1}) \langle D \rangle; \quad \langle \diagdown \rangle = \langle \diagup \rangle - q \langle \bigcirc \rangle$$

If D has n_+ positive crossings, n_- negative crossings, then the Jones polynomial of the underlying link L is

$$J(L) = (-1)^{n_-} q^{n_+ - 2n_-} \langle D \rangle$$

Knot / 3-manifold invt., Pre-Khovanov

Quantum	Gauge Theoretic
Jones, HOMFLY, WRT...	Instanton Floer Homology Casson Invariant...



The Casson invt. is twice the Euler char. of Instanton Floer.

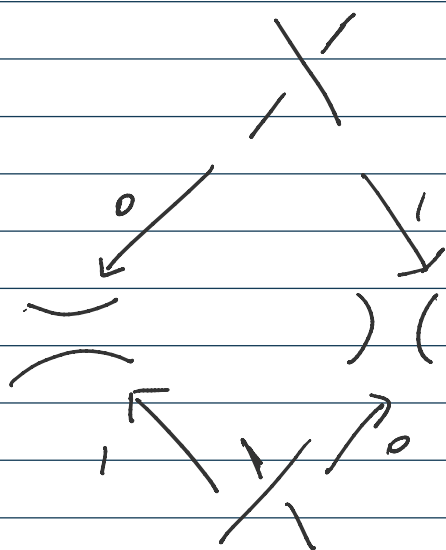
Khovanov homology

link (diagram) $L \rightsquigarrow$ bigraded $\mathbb{Z}$ -module $Kh^{*,*}(L)$

s.t. $J(L) = \sum_{i,j} (-1)^i q^j \text{rk}_{\mathbb{Z}} Kh^{i,j}(L)$

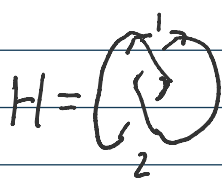
Jones polynomial from cube of resolutions

Given D , order its crossings. For each crossing we have a 0-res. and a 1-res.

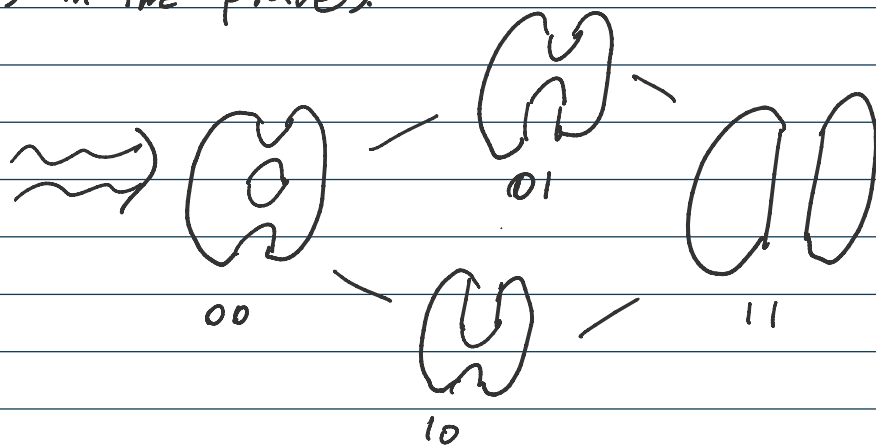


If $n = n_+ + n_-$, every elt. of $\{0,1\}^n$ gets a complete resolution (collection of circles in the plane).

E.g.



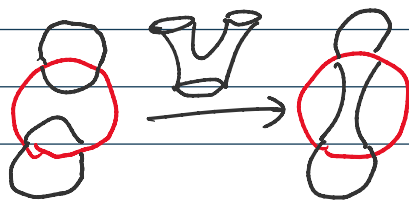
$n_+ = 0, n_- = 2$



$$\begin{aligned}\langle H \rangle &= (q+q^{-1})^2 - q(q+q^{-1}) - q(q+q^{-1}) + q^2(q+q^{-1}) \\ &= q^4 + q^2 + 1 + q^{-2}\end{aligned}$$

$$J(H) = (-1)^2 q^{-4} \langle H \rangle = 1 + q^{-2} + q^{-4} + q^{-6}$$

Changing a 0-res. to a 1-res. is given by a saddle.

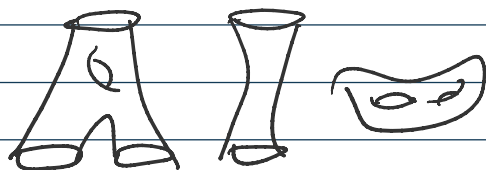


We will interpret the Khovanov cx as a cx in a category of pictures.

Let $\overline{\text{Cob}}^3$ be the category with

Objects: Collections of oriented circles in $\mathbb{R}^2$

Morphisms: Oriented cobordisms in $\mathbb{R}^2 \times [0, 1]$



For $\Sigma \in \text{Mor}(\mathcal{O}_1, \mathcal{O}_2)$, define $\deg \Sigma = \chi(\Sigma)$,

Add objects $\mathcal{O}\{m\}$ for each $\mathcal{O} \in \text{Obj}(\overline{\text{Cob}}^3) \leadsto$ Graded cat. Cob^3
 $m \in \mathbb{Z}$

Let $\text{Mat}(\text{Cob}^3)$ denote the additive closure of Cob^3

$\text{Obj} =$ formal direct sums of objects in Cob^3

$\text{Mor} =$ matrices of $\mathbb{Z}$ -linear combinations of morphisms.

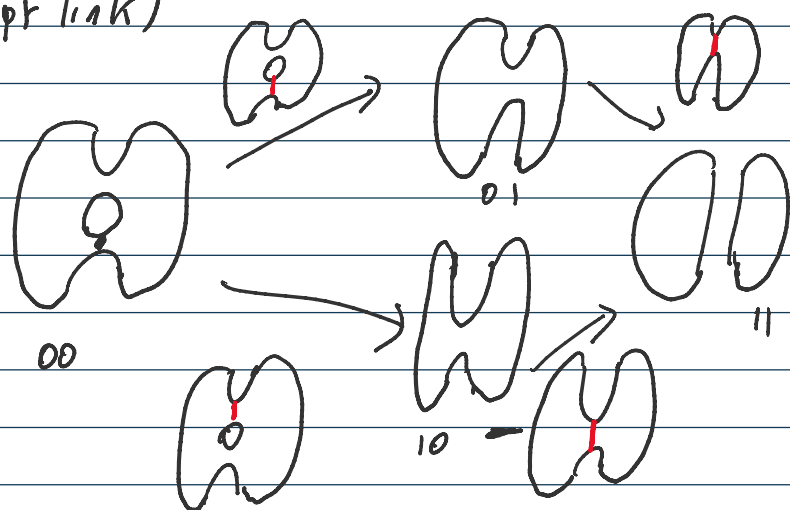
Obj = normal direct sums of objects in Cob^3
 Mor = matrices of $\mathbb{Z}$ -linear combinations of morphisms.

$\text{Kom}(\text{Mat}(\text{Cob}^3)) = \text{Cat. of chain complexes over } \text{Mat}(\text{Cob}^3).$

For $\alpha \in \{0, 1\}^n$, assign the object $\mathcal{O}_{\sum \alpha_i}$, $\mathcal{O}$ is the res. of L gotten from the α_i : $\sim$ res. at the i^{th} crossing, $m = \sum \alpha_i$.

For each edge, assign the saddle cobordism b/w objects at each edge.

(Hopf link)



$d^2 = 0$ mean that all square faces in the diagram anti-commute

Signs: Add signs to the edges s.t. each square face gets an odd # of minus signs.

, lives $\text{Kom}(\text{Mat}(\text{Cob}^3))$

The m^{th} chain gp. $[L]^m$ of $[L]$ is $\bigoplus_{\alpha \in \{0,1\}^n: \sum \alpha_i = m} \bigoplus$ (resolution over α)

The chain htpy type of $[L]$ is a link invariant after passing to a quotient of Cob^3 .

$\text{Cob}^3_{\text{rel}}$ is Cob^3 mod local relations on morphisms:

$$1 \cdot P, P^2 \quad 1 \cdot A, A^2 \quad 1 \cdot AP, AP^2 \quad 1 \cdot AA$$

$Cob^3_{\mathbb{Z}}$ is Cob^3 mod local relations on morphisms:

$$\bigcirc_T = 2, \quad \bigcirc_S = 0, \quad \text{and} \quad 4TU = 0$$

Bar-Natan category $BN := Kom(Mat(Cob^3_{\mathbb{Z}}))$.

$$\cap \longleftrightarrow \cup$$

Proof of Reidemeister I invariance: $\searrow \cup = (\cap) \circ \rightarrow (\searrow)$

$$\begin{array}{ccc} \bigcirc & \xrightarrow{0} & 0 \\ \uparrow & & \uparrow \\ F^0 & & G^1 = 0 \\ \downarrow & & \downarrow \\ \bigcirc \cup & \xrightarrow{d} & \bigcirc \cap \end{array}$$

h is the chain htpy from $F \circ G$ to id
 $0 = F^1$ to id

$$h = \text{diagram}$$

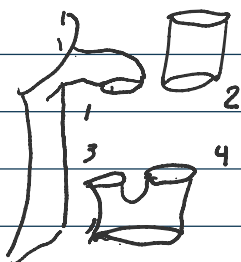
$$dF^0 = \text{diagram} = 0, \quad G^0 F^0 = \text{diagram} = id$$



Clearly $\underbrace{F'G'}_{=0} - \text{id} + dh = 0.$

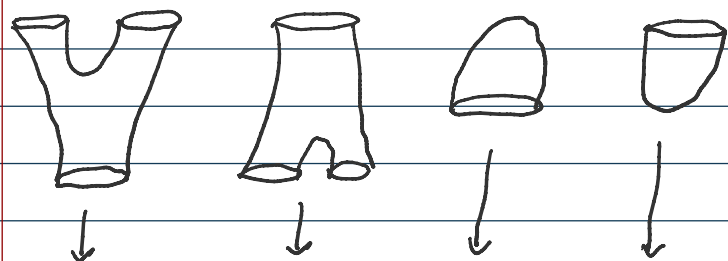
$$F^0 G^0 - \text{id} + dh = \underbrace{\text{diagram 1}}_{=0} - \underbrace{\text{diagram 2}}_{=F^0 G^0} - \text{diagram 3} + \text{diagram 4}$$

by 4T₀ applied to



Any functor from $\text{Cob}^3_{\mathbb{R}} \rightarrow \text{Abelian Category}$ gives a knot invariant.

Any cobordism in $\text{Mor}(\text{Cob}^3_{\mathbb{R}})$ can be decomposed into:



by Morse theory.

(For $\mathcal{F}$ a TRFT) $\mathcal{F}(S') = V$

$$\mathcal{F} = \begin{matrix} \Delta: V \rightarrow V^{\otimes 2} \\ m: V \otimes V \rightarrow V \\ l: V \rightarrow \mathbb{Z} \\ \epsilon: V \rightarrow \mathbb{Z} \end{matrix}$$



$$\mathcal{F}(\sqcup_k 0) = V^{\otimes k} ; \quad V = \mathbb{Z}\{\cdot 1\} \oplus \mathbb{Z}\{1\cdot\} \quad \square \simeq \square$$

$$F(\sqcup_k \bigcirc) = V^{\otimes k} ; \quad V = \mathbb{Z}\langle -1 \rangle \oplus \mathbb{Z}\langle 1 \rangle \quad \boxed{\bigcirc} \approx \boxed{\bigcirc}$$

$$m = \begin{cases} v_- \otimes v_- \mapsto 0, & v_- \otimes v_+ \mapsto v_- \\ v_+ \otimes v_+ \mapsto v_+, & v_+ \otimes v_- \mapsto v_- \end{cases} \quad l = \begin{cases} 1 \mapsto v_+ \end{cases}$$

$$\Delta = \begin{cases} v_- \mapsto v_- \otimes v_- \\ v_+ \mapsto v_+ \otimes v_- + v_- \otimes v_+ \end{cases} \quad \varepsilon = \begin{cases} v_- \mapsto 1 \\ v_+ \mapsto 0 \end{cases}$$

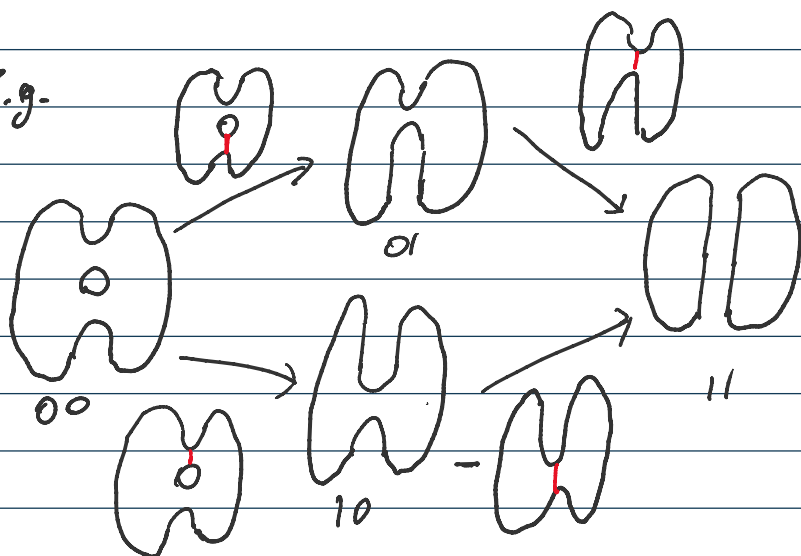
The grading on V is the Jones grading

called

$$\text{Define } Kh(L) = H^*(F(\llbracket L \rrbracket) [-n_-] \langle n_+ - 2n_- \rangle)$$

$$\text{Thm: } J(L) = \sum_{i,j} (-1)^{i+j} \text{rk}_{\mathbb{Z}} Kh^{i,j}(L).$$

E.g.



$$(V \otimes V)\langle -4 \rangle \xrightarrow{m \oplus m} V\langle -3 \rangle \oplus V\langle -3 \rangle \xrightarrow{\Delta' - \Delta^2} \underline{(V \otimes V)\langle -2 \rangle}$$

Functoriality: Cobordisms in $\mathbb{R}^3 \times [0, 1]$ decompose into local moves

$$\cap \leftrightarrow \Lambda \leftrightarrow \cup, \quad \text{I} \leftrightarrow \text{II}$$

$$\text{III} \leftrightarrow \text{IV} \quad \text{RI, II, III}$$

$$\bigcirc \leftrightarrow \emptyset \quad \text{cup/cap} \quad \text{) (} \leftrightarrow \text{ } \cup \cup$$

saddle

Thm (Jacobsson, Bar-Natan) The map on $[[L]]$ associated to a cobordism in $\mathbb{R}^3 \times [0, 1]$ only is independent of its decomposition into elementary cobordisms up to a sign and chain homotopy in the Bar-Natan Category.

